Jeff Mutuku

Human Performance Researcher | Exercise Physiology | Sports Science | Data Analytics

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PROFESSIONAL PROFILE

I work at the intersection of exercise physiology, collegiate sports performance, and data analysis. My experience spans metabolic and body-composition assessment, NCAA Division I performance support, and collaborative human-subject research. I am interested in research and R&D opportunities that bring rigorous measurement and clear scientific communication into practice.

PROFESSIONAL EXPERIENCE

Performance Scientist | Wellth Performance

May 2026 - Present

Conduct metabolic assessments including resting metabolic rate and maximal exercise testing, alongside DEXA and 3D body-composition assessments. Develop AI-assisted metabolic interpretation and reporting workflows and contribute to body-composition research and analytics.

Indirect calorimetry · DEXA · Physiological reporting

Performance Coach Intern | California Baptist University · Athletic Performance Center

July 2025 - May 2026

Supported performance operations across 12 NCAA Division I athletic programs. Conducted countermovement jump assessments and supported workload monitoring with STATSports GPS and accelerometry alongside performance staff and coaches.

Force plates · CMJ testing · STATSports GPS

EDUCATION

M.S. in Kinesiology | California Baptist University

May 2024 - August 2026 | Concentration: Exercise Science | GPA 3.89 / 4.00

B.A. in Sports Management | Notre Dame College

Fall 2019 - December 2022 | Cum Laude · Dean's List | GPA 3.70 / 4.00

SELECTED COLLABORATIVE RESEARCH

Warm-up strategies & sprint swimming

Exploring acute responses to upper-body, lower-body, and combined conditioning activities in collegiate swimmers.

Applied testing and dashboard development, as described in the resume and project documentation.

Status: Study & dashboard documented.

Seasonal heart-rate recovery

Comparing post-exercise recovery kinetics in Division I swimmers during in-season and off-season testing.

Named poster coauthor (Mutuku J). The resume describes testing and Python recovery modeling.

Status: Research poster completed.

BFR walking & cardiometabolic markers

A four-week study of low-frequency blood-flow-restriction walking in overweight and obese adults.

Named poster coauthor (Mutuku J). Individual study responsibilities are not specified in the supplied poster.

Status: Research poster completed.

Research projects and poster artifacts are not presented as peer-reviewed publications. See portfolio case studies for methods, findings and limitations.

METHODS & TECHNOLOGIES

Physiological assessment: VO2max and maximal exercise testing, resting metabolic rate, DEXA, 3D body scanning

Research methods: Human-subject testing, protocol execution, participant screening, informed consent, data quality and scientific communication

Biomechanics & monitoring: Countermovement jump assessment, ground reaction force, impulse, accelerometry and GPS workload monitoring

Analysis & development: Python, Pandas, NumPy, SciPy, exponential modeling, Plotly, HTML, CSS and JavaScript

Reporting tools: Excel, Power BI, Streamlit, Git/GitHub and Vercel

CREDENTIALS & PROFESSIONAL DEVELOPMENT

NASM Certified Personal Trainer

California Department of Public Health X-Ray Technician Permit

STATSports Certified GPS User

FIFA Diploma in Football Medicine

NSCA member · CPSS exam candidate (2026). CPSS is exam candidacy, not an earned certification.